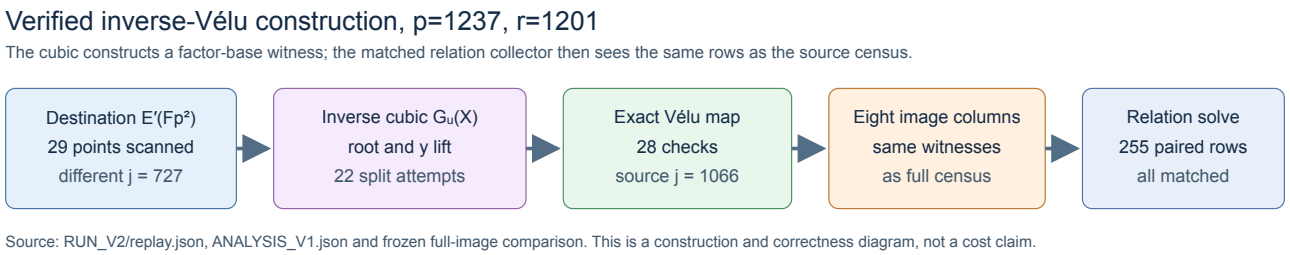
Order-1,201 inverse-Vélu factor-base result

Frozen same-target native comparison; three rounds, five arms.

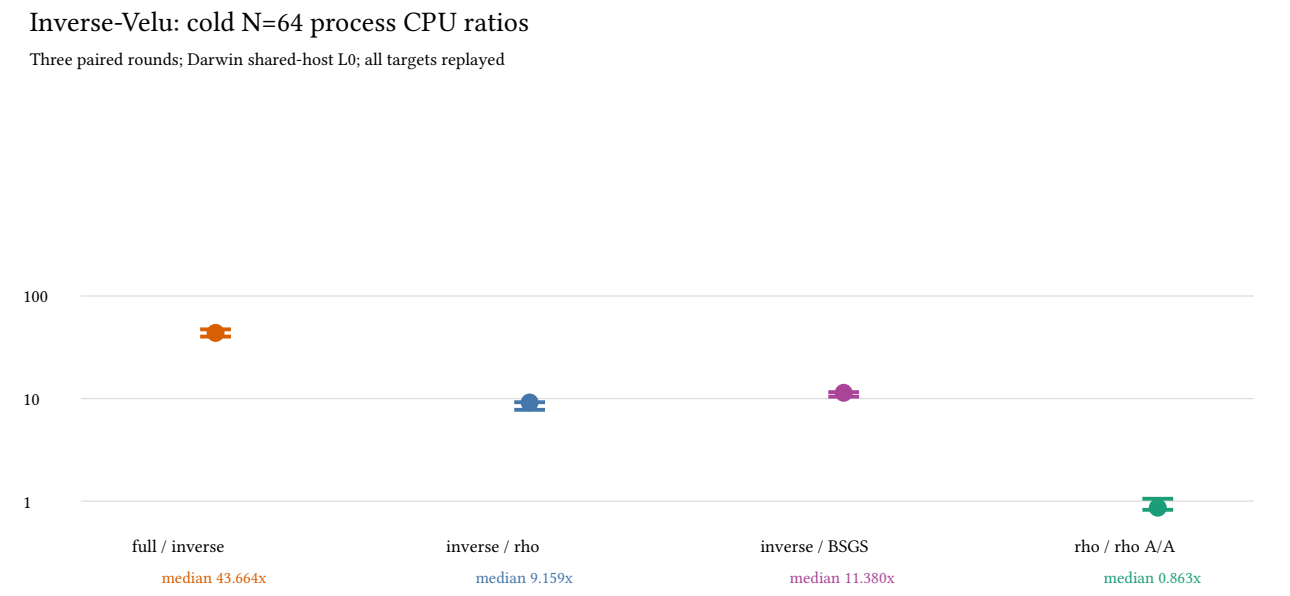
The degree-three inverse x-coordinate cubic over F_{1237^2} constructed the same eight image-base columns as the full source census after 29 destination-point checks. Every one of 255 paired inverse/full target relation rows matched. The independent checker replayed 1,275 scalar outputs and 6,126 relation witnesses across 60 successful cold child processes.



The single counted unit in this table is $S_{group}=(additions+doublings)/(N\sqrt{1201})$. It includes the complete child through final scalar replay, but is a lower bound on fully priced portable S.

arm	cold N=1	reuse N=64
inverse-Vélu plus relations	149.645	151.595
full-image census plus relations	88,514.859	1,532.302
source BSGS	2.972	1.315
source signed rho	5.107	4.071
rho A/A	5.107	4.071

At N=64, paired group full/inverse was 10.108, inverse/rho 37.235, and inverse/BSGS 115.266. The corresponding paired field-multiply and field-inversion inverse/rho ratios were 41.699 and 37.694. Whole-child CPU ratios on the shared Darwin host were 43.664, 9.159 and 11.380 for full/inverse, inverse/rho and inverse/BSGS; rho A/A spanned 0.823–1.054. Those CPU values are L0 descriptive.



The old construction census used 3,063,003 counted group calls in round zero; direct inverse construction used 667. Its relation yield remained identical, so the algorithmic next test is a separate nonfactorable `chi` decomposition polynomial oracle with exhaustive pair-enumeration control and a complete target solve. Disjoint native pricing, L2 timing and a broader subgroup-order trend remain open. `RESULTS.md` links the frozen spec, raw run, independent replay, Lean identity scope and follow-on protocol.